

Certified exact sums of primes over dyadic intervals via a ladder of generalized factorials

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Abstract

For an integer $m \geq 8$ let $S(m) = \sum_{m < p \leq 2m} p$, the sum over primes in the dyadic window $(m, 2m]$. We construct a ladder of generalized factorials $A_j(2m) = \prod_{i \leq 2m} i^{Q_j(i)}$ whose integer exponent sequences $Q_j(i) = m^j T_j(\frac{2i-3m}{m})$ are Chebyshev polynomials rescaled to take integer values. The p -adic valuation of $A_j(2m)$ at a prime $p \in (m, 2m]$ equals $Q_j(p)$, so each rung of the ladder yields one exact linear functional (“moment”) of the interval’s primes, computable from the primes $\leq m$ together with logarithms of the integers $\leq 2m$, without identifying any prime inside the interval. A near-minimax Chebyshev combination of $K+1$ moments approximates $t/\log t$; a rigorous sandwich, weighted by the zeroth moment, encloses $S(m)$ in an interval of certified width. Since $S(m)$ is an integer, a certified width below 1 determines it exactly. An implementation in 260-bit interval arithmetic certifies $S(10^3) = 200,923$, $S(10^4) = 15,434,795$ and $S(10^5) = 1,255,204,276$ using 6, 8 and 10 moments respectively, with empirical window decay of roughly one order of magnitude per additional moment, above the proven geometric floor $3 + 2\sqrt{2}$. We situate the construction relative to Chebyshev-type identities and to combinatorial prime-summing algorithms, and we state its limitations plainly: the method is a certified exact procedure of logarithmic description length, not an asymptotically fast algorithm and not a closed form.

Keywords: primes in dyadic intervals; Chebyshev’s method; generalized factorials; p -adic valuations; Chebyshev polynomials; certified computation; interval arithmetic.

MSC 2020: primary 11N05; secondary 11Y35, 41A50, 65G30.

1 Introduction

Exact computation of prime-counting and prime-summing functions has two established traditions. The combinatorial tradition descends from Meissel through Lehmer, Lagarias–Miller–Odlyzko and Deléglise–Rivat, and computes $\pi(x)$ in $O(x^{2/3} \log^{-2} x)$ time [2, 3]; it has been extended to the prime power sums $\sum_{p \leq x} p^k$ [4], with widely used implementations [5], and has a folklore branch, the $O(x^{3/4})$ dynamic program posted by the Project Euler user Lucy_Hedgehog in 2013, which sums the primes below x “without finding them all” [6]. The analytic tradition,

*Version 1.0, August 2026. Companion to *Prime counts and prime sums from multiplication tables: a self-similar formula system* [33] (same author). Developed in collaboration with Claude (Anthropic); all numerical claims were machine-verified as described in Section 5. Responsibility for the content rests with the author. A full disclosure of AI use appears at the end of the paper.

initiated by Lagarias and Odlyzko [16] and realized rigorously by Platt [7] and Büthe [8], evaluates an explicit formula over zeta zeros to accuracy below $\frac{1}{2}$ and rounds; the target being an integer converts a certified approximation into an exact determination — a closure used likewise, outside prime counting, in Johansson’s certified computation of the partition numbers [18] — and rigorous computations of the Chebyshev function θ itself belong to the same tradition [17]. The need for certification is not hypothetical: a 2009 computation of the sum of the primes below 10^{21} was withdrawn in 2011 after an off-by-one discrepancy came to light [5].

This note describes a third route to exact interval prime sums, elementary in its ingredients and certified in its execution. It originates in Chebyshev’s 1852 observation [1] that the prime factorization of the central binomial coefficient controls $\theta(2m) - \theta(m)$: the primes of $(m, 2m]$ appear in $\binom{2m}{m}$ with exponent exactly one. We extend this single exact functional to a *ladder* of them. The factorial is the weight-1 member of the family $\prod_{i \leq n} i^{w(i)}$; taking polynomial weights w produces generalized factorials¹ (the weight $w(i) = i$ gives the classical hyperfactorial [9, 10]) whose valuations at large primes are $w(p)$ itself. Choosing the weights to be integer-rescaled Chebyshev polynomials yields a numerically stable basis of exact moments; approximation theory then converts finitely many moments into a rigorous two-sided enclosure of $S(m)$, and the integrality of $S(m)$ converts a sufficiently narrow enclosure into the exact value.

We make three claims and two explicit non-claims. Claimed: (i) the exact extraction identities of Section 2; (ii) the certified enclosure protocol of Sections 3–4, whose window width contracts at a guaranteed geometric rate in the number of moments; (iii) certified computations at three scales, verified against sieves. Not claimed: computational superiority over the combinatorial algorithms, which remain asymptotically faster; and novelty of the individual ingredients, each of which is classical. To the best of our knowledge the assembly — polynomial-weight generalized factorials organized as a moment ladder and closed into a certified integer determination — has not appeared in the literature, but we regard this as a statement about our searches, not a theorem.

2 The ladder and its exact moments

Fix an integer $m \geq 8$ and write $n = 2m$, $u(t) = (2t - 3m)/m$, so that u maps $[m, 2m]$ onto $[-1, 1]$. Let T_j denote the Chebyshev polynomials of the first kind and define integer sequences

$$Q_j(i) = m^j T_j(u(i)), \quad j \geq 0.$$

That $Q_j(i) \in \mathbb{Z}$ for $i \in \mathbb{Z}$ follows from the integer recurrence $Q_0 = 1$, $Q_1(i) = 2i - 3m$, $Q_{j+1}(i) = 2(2i - 3m)Q_j(i) - m^2Q_{j-1}(i)$. Define the j -th ladder object

$$A_j(n) = \prod_{i=1}^n i^{Q_j(i)} \in \mathbb{Q}_{>0},$$

a rational number in general since Q_j takes both signs; $A_0(n) = n!$ and, for the weight i in place of Q_1 , one obtains the hyperfactorial. The ladder interacts with primes through the following elementary facts.

¹The term is overloaded: Bhargava’s generalized factorials [26] generalize $n!$ to subsets of $\mathbb{Z}$ via p -orderings, a different construction; throughout this paper “generalized factorial” means a product $\prod_{i \leq n} i^{w(i)}$ with polynomial weight w .

Lemma 1 (Interval valuations). *Let p be prime with $m < p \leq 2m$. Then $v_p(A_j(n)) = Q_j(p)$.*

Proof. Since $p > m = n/2$, the only multiple of p in $[1, n]$ is p itself, and $p^2 > n$ for $m \geq 2$. Hence $v_p(A_j(n)) = \sum_{i \leq n} Q_j(i) v_p(i) = Q_j(p)$. $\square$

Lemma 2 (Below-interval valuations). *Let $p \leq m$ be prime. Then*

$$v_p(A_j(n)) = \sum_{a \geq 1, p^a \leq n} \sum_{t=1}^{\lfloor n/p^a \rfloor} Q_j(p^a t),$$

and each inner sum is an exact integer computable in $O(j^2)$ arithmetic operations from the coefficient list of Q_j , via the exact power sums $\sum_{t=1}^T t^r = \sum_{q=1}^r q! S(r, q) \binom{T+1}{q+1}$, where $S(r, q)$ are Stirling numbers of the second kind.

Proof. The double sum is $\sum_{i \leq n} Q_j(i) v_p(i)$ rewritten by counting, for each prime power p^a , the indices i it divides. Substituting $i = p^a t$ makes the inner sum a polynomial in t summed over an initial segment, which the stated Faulhaber-type identity evaluates exactly. $\square$

Proposition 1 (Exact moment extraction). *For every $j \geq 0$,*

$$M_j := \sum_{m < p \leq 2m} Q_j(p) \log p = \sum_{i=2}^n Q_j(i) \log i - \sum_{p \leq m} v_p(A_j(n)) \log p,$$

with the valuations given exactly by Lemma 2. The right-hand side involves only the primes $\leq m$ and logarithms of integers $\leq n$; no integer in $(m, 2m]$ is examined for primality.

Proof. Take logarithms in the prime factorization of $A_j(n)$, split the primes at m , and apply Lemma 1 to the upper range. Prime powers $p^a \leq n$ with $a \geq 2$ have $p \leq \sqrt{n} \leq m$ and are absorbed by Lemma 2. $\square$

The case $j = 0$ is Chebyshev's identity in the form $\theta(2m) - \theta(m) = \log n! - \sum_{p \leq m} v_p(n!) \log p$; we write $\mu_0 = M_0$.

3 From moments to the integer

Let $f(t) = t/\log t$, so that $f(p) \log p = p$ and $S(m) = \sum_{m < p \leq 2m} f(p) \log p$. Fix coefficients $a_0, \dots, a_K \in \mathbb{R}$ (in practice, dyadic rationals), set $P(t) = \sum_{j \leq K} a_j T_j(u(t))$ and $g = f - P$ on $[m, 2m]$. Since $Q_j(p) = m^j T_j(u(p))$,

$$S(m) = \sum_{j=0}^K a_j m^{-j} M_j + \sum_{m < p \leq 2m} g(p) \log p, \quad \mu_0 \min_{[m, 2m]} g \leq \sum_p g(p) \log p \leq \mu_0 \max_{[m, 2m]} g, \quad (1)$$

the bounds valid because every $\log p > 0$ and $\sum_p \log p = \mu_0$. Thus any choice of P yields a rigorous enclosure of $S(m)$ of width $\mu_0 \cdot \text{osc}_{[m, 2m]}(g)$; when the certified width is below 1, the unique integer in the enclosure is $S(m)$. Validity does not depend on optimality of P , which may therefore be designed in floating point and then frozen.

Two elementary bounds certify the range of g from finitely many evaluations.

Lemma 3 (Curvature bounds). *For $t \geq e^2$ one has $|f''(t)| = |\frac{2}{\log t} - 1|/(t \log^2 t) \leq 1/(m \log^2 m)$ on $[m, 2m]$, and $\max_{[-1,1]} |T_j''| = j^2(j^2 - 1)/3$. Consequently $\|g''\|_\infty \leq B_2 := \frac{1}{m \log^2 m} + \frac{4}{m^2} \sum_{j \leq K} |a_j| \frac{j^2(j^2 - 1)}{3}$, and for a uniform grid of step s containing both endpoints,*

$$\max_{[m, 2m]} g \leq \max_i g(t_i) + \frac{s^2}{8} B_2, \quad \min_{[m, 2m]} g \geq \min_i g(t_i) - \frac{s^2}{8} B_2.$$

Proposition 2 (Geometric window decay). *Let E_K denote the error of best uniform approximation to f on $[m, 2m]$ by polynomials of degree K . Then $\limsup_{K \rightarrow \infty} E_K^{1/K} \leq 1/\rho_m$, where ρ_m is the Bernstein-ellipse parameter of the nearest singularity of f ($t = 1$, where $\log t = 0$) relative to $[m, 2m]$; as $m \rightarrow \infty$, $\rho_m \rightarrow 3 + 2\sqrt{2} \approx 5.83$. Hence the achievable certified window contracts at least geometrically in K , and the number of moments needed to reach width 1 grows only logarithmically in the initial window.*

Proof sketch. f is analytic in a neighborhood of $[m, 2m]$ with nearest singularities at $t = 1$ and $t = 0$; under $u(t)$ these map near $u = -3$, and $\rho = |u_*| + \sqrt{u_*^2 - 1}$ with $u_* \rightarrow -3$. The bound is then the classical Bernstein estimate; see [15, Ch. 8]. The limiting constant is itself classical: $3 + 2\sqrt{2} = (\sqrt{2} + 1)^2 = (\sqrt{\kappa} + 1)/(\sqrt{\kappa} - 1)$ at endpoint ratio $\kappa = 2$, the familiar Chebyshev rate for intervals of condition ratio 2, known since Chebyshev’s best uniform approximation of the reciprocal [22]. $\square$

Remark 1. Empirically the contraction factor observed in Section 5 is close to 10 per moment, faster than the guaranteed floor; the floor alone already gives the logarithmic moment count.

Remark 2 (Moment-problem reading). The sandwich (1) is the elementary end of a classical subject: two-sided bounds on a linear functional of a positive measure from finitely many polynomial moments belong to the Chebyshev–Markov–Stieltjes circle [23, 24]. Gauss-quadrature-type bounds [25] would use all $K+1$ moments inside the bound itself, conceivably tightening the window at fixed K ; we have not pursued this.

4 Certified implementation

The protocol has four stages (reference implementation in the code repository [34]), all executed in ball (interval) arithmetic at 260 bits except where exact integer arithmetic is available. (1) *Moments.* The anchor $\sum_{i \leq n} Q_j(i) \log i$ is accumulated with $Q_j(i)$ produced by the integer recurrence and $\log i$ as an interval; the corrections of Lemma 2 are exact integers multiplied by interval logarithms. This yields enclosures of $M_0, \dots, M_K$ whose observed widths at the scales below were at most 4.2×10^{-11} (and as small as 10^{-71}). (2) *Design.* $a_0, \dots, a_K$ are Chebyshev interpolation coefficients of f computed in floating point and frozen exactly as dyadics. (3) *Residual certification.* g is evaluated in interval arithmetic on an integer grid of step s and extended to $[m, 2m]$ by Lemma 3. (4) *Assembly.* Equation (1) is evaluated in interval arithmetic; if $\lceil \log \rceil = \lfloor \log \rfloor$, that integer is $S(m)$.

The trust base is: correctness of the interval arithmetic library — mpmath [19], whose interval context is documented as experimental; the pipeline therefore touches only interval addition, subtraction, multiplication, division, and the logarithm, and a port to ball arithmetic in Arb [20] would further harden this leg — of big-integer arithmetic, and of Lemma 3’s two analytic inequalities. All other quantities are enclosed, not estimated; for general background on validated numerics see [21]. Ground-truth verification by segmented sieve, and cross-checks

of every moment enclosure against direct computation over the (sieved) interval primes, are performed outside the pipeline.

5 Results

window	moments $K+1$	certified width	$S(m)$ determined	sieve check
$(10^3, 2 \cdot 10^3]$	6	0.140	200,923	agrees
$(10^4, 2 \cdot 10^4]$	8	0.155	15,434,795	agrees
$(10^5, 2 \cdot 10^5]$	10	0.253	1,255,204,276	agrees

Runtimes on commodity hardware were 0.3, 3.1 and 31 seconds respectively, dominated by the $O(m)$ interval logarithms of the anchors. Measured window contraction per added moment was 9.7–11.4 across the three scales. In uncertified experiments the same estimator tracked sieved interval sums with relative errors of order 10^{-4} at additional scales.

6 Related identities and remarks

Rung 0 is Chebyshev’s binomial identity [1]. The weight- i rung is governed by the hyperfactorial, whose p -adic valuations have an independent literature [9, 10]; the exponent formula involves triangular numbers, and more generally the corrections of Lemma 2 are the Faulhaber closed forms of iterated-sum sequences, so the ladder may be read as a multiplicative avatar of the tower $1, n, \binom{n+1}{2}, \binom{n+2}{3}, \dots$. High-precision evaluation of the anchors by Euler–Maclaurin summation leads to the generalized Glaisher–Kinkelin (Bendersky–Adamchik) constants, expressible through $\zeta'(-k)$ [11, 12, 13]; we have not needed this refinement at the scales reported, but it removes the $O(m)$ anchor cost in principle. A prime-free variant of the zeroth-order theory is provided by Linnik’s identity [14], which expresses $\Lambda(n)/\log n$ as an alternating-harmonic combination of the counts of ordered factorizations into parts ≥ 2 , and was developed into prime-counting and prime-power-sum algorithms by McKenzie [32]; the companion paper [33] builds this route into a uniform weighted formula system which, in exact rational arithmetic, reproduces $\sum_{p \leq 10^5} p = 454,396,537$ from divisor-lattice data alone.

The nearest structural relatives of the ladder that our searches found are the partial factorizations of products of binomial coefficients studied by Lagarias, Mehta, and Du [27, 28]: the same threshold-split valuations of factorial-like products, analyzed there asymptotically (under the prime number theorem or the Riemann hypothesis) rather than assembled into exact window functionals. The Chebyshev–Sylvester weight schemes, from Sylvester through Diamond–Erdős [29] to a recent computational revival [30], optimize finitely supported weights acting on $\log[x]!$ into one-sided bounds on ψ — inequalities of increasing sharpness, never exact functionals of a window; and a family of interval-count identities extracted from binomial-coefficient valuations, with $O(\sqrt{k})$ errors, appears in [31]. Finally, the dyadic windows $(2^{r-1}, 2^r]$ tile the integers and each contains a prime by Bertrand’s postulate, so the pair $(I(r), F)$ with $I(r) = (2^{r-1}, 2^r]$ and F the certified procedure above answers, for every r , the question “what is the sum of the primes in the r -th dyadic window” exactly.

7 Limitations and open questions

The method as implemented costs $O(m)$ interval logarithms and is therefore not competitive with the $O(x^{2/3})$ combinatorial algorithms [2, 4] as a computational tool; its interest is methodological. The corrections consume the primes below the window; a fully self-contained variant recurses dyadically to a trivial base, or replaces rung 0 by the Linnik route [32, 33]. Three questions seem worth recording. (1) Can the anchors be evaluated to certified accuracy in time polylogarithmic in m via the Euler–Maclaurin/ $\zeta'(-k)$ route, making the whole pipeline sublinear? (2) Does the ladder yield explicit *inequalities* for $S(m)$ of Chebyshev strength but higher order — bounds rather than computations — that the combinatorial algorithms do not provide? (3) Is there a principled obstruction theorem formalizing the folklore that no fixed-length elementary closed form for $S(m)$ exists, given that the fluctuating part of $S(m)$ carries the zeta spectrum? We do not know, and we would welcome pointers to prior appearances of this construction.

Disclosure of AI use

In accordance with publisher policies on the use of artificial intelligence, the author discloses the following. This paper was developed in an extensive collaboration with Claude (Anthropic; Claude Fable 5, accessed through the `claude.ai` interface, 2026). The direction of the work — the research programme of exact moment systems for interval primes — together with all decisions about scope, claims, and submission, are the author’s. Within that direction, the AI system contributed substantially to the mathematical development and proofs, wrote the verification code, drafted and revised the manuscript text, and assisted with literature search; the reason for its use was to enable a single non-specialist author to develop, check, and document the material to a publishable standard. Every numerical and mathematical claim was subsequently machine-verified by the certified computations reported in Section 5, with the verification code openly available in the cited repository [34]. The author has reviewed the entire manuscript and takes full responsibility for its content, including the accuracy of the mathematics and of all references.

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